

Neural Networks Explained

Neural networks are computational models inspired loosely by the structure of the human brain, consisting of layers of interconnected nodes called neurons. Each connection between neurons carries a weight that is adjusted during training to minimize the difference between the network's predictions and the true outcomes. A basic neural network contains an input layer, one or more hidden layers, and an output layer. Information flows forward through the network, with each neuron applying a mathematical function called an activation function to introduce non-linearity, allowing the network to learn complex patterns. Training a neural network typically involves an algorithm called backpropagation, which calculates how much each weight contributed to the overall error and adjusts the weights accordingly using gradient descent. Deep learning refers to neural networks with many hidden layers, which have proven remarkably effective at tasks such as image recognition, speech processing, and natural language understanding. Convolutional neural networks, a specialized architecture, excel at processing grid-like data such as images by applying filters that detect edges, textures, and shapes. Recurrent neural networks and their more advanced variant, long short-term memory networks, are designed to handle sequential data such as text or time series, since they maintain a form of memory across time steps. More recently, transformer architectures have largely replaced recurrent networks in language tasks due to their ability to process entire sequences in parallel while capturing long-range dependencies through a mechanism called attention.